

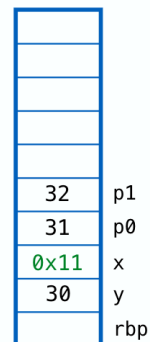
Worksheet 8B

Quiz 1: Garbage in the middle

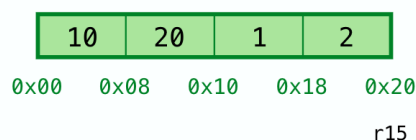
Which cells are garbage?

- a) 0x00 and 0x08
- b) 0x08 and 0x10
- c) 0x18 and 0x20
- d) None
- e) All

```
let y = let tmp = (10, 20)
      in tmp[0] + tmp[1]
, x = (1, 2)
, p0 = x[0] + y
, p1 = x[1] + y
in (p0, p1)
```



Lets reclaim & recycle garbage!



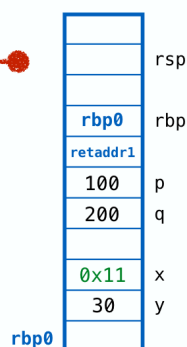
Quiz 2: Garbage in the middle with Stack

Which cells are garbage?

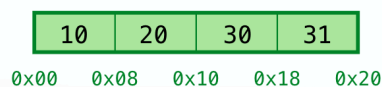
- a) 0x00 and 0x08
- b) 0x08 and 0x10
- c) 0x10 and 0x18
- d) None
- e) All

```
def foo(p, q):
    let tmp = (p, q)
    in tmp[0] + tmp[1]

let y = foo(10, 20)
, x = (y, y + 1)
, z = foo(100, 200)
in x[0] + z
```



Lets reclaim & recycle garbage!



Quiz 3: Recursive Data

What is the heap when range(0, 3) returns?

```
def range(i, j):
    if (j <= i): false else: (i, range(i+1, j))

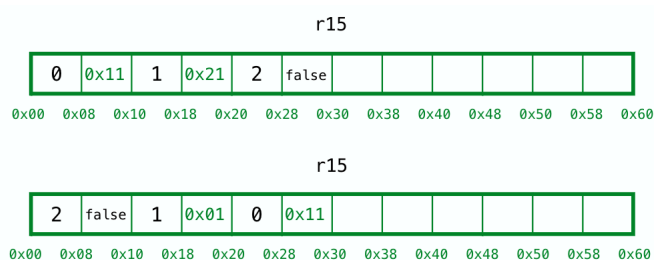
def sum(l):
    if l == false: 0 else: l[0] + sum(l[1])

let t1 =
    let l1 = range(0, 3)
    in sum(l1)
, l = range(t1, t1 + 3)
in (1000, l)
```



(A)

(B)



Quiz 4: Recursive Data

```
def range(i, j):
    if (j <= i): false else: (i, range(i+1, j))

def sum(l):
    if l == false: 0 else: l[0] + sum(l[1])

let t1 =
  let l1 = range(0, 3)
  in sum(l1)
, l = range(t1, t1 + 3)
in
  (1000, l)
```

call range(3,6)

2	false	1	0x01	0	0x11	5	false	4	0x31	3	0x41
---	-------	---	------	---	------	---	-------	---	------	---	------

0x00 0x08 0x10 0x18 0x20 0x28 0x30 0x38 0x40 0x48 0x50 0x58 0x60

What is the value of `l` at the point indicated by the red arrow?

- a) 0x30
- b) 0x31
- c) 0x50
- d) 0x51
- e) 0x60

Quiz 5: Live Cells

QUIZ: Which cells are “live” on the heap?

(A) 0x00
(B) 0x10
(C) 0x20
(D) 0x30
(E) 0x40
(F) 0x50

2	false	1	0x01	0	0x11	5	false	4	0x31	3	0x41
---	-------	---	------	---	------	---	-------	---	------	---	------

0x00 0x08 0x10 0x18 0x20 0x28 0x30 0x38 0x40 0x48 0x50 0x58 0x60

QUIZ: Which cells are “live” on the heap?

(A) 0x00
(B) 0x10
(C) 0x20
(D) 0x30
(E) 0x40
(F) 0x50

2	0x51	1	0x01	0	0x11	5	false	4	0x31	3	0x41
---	------	---	------	---	------	---	-------	---	------	---	------

0x00 0x08 0x10 0x18 0x20 0x28 0x30 0x38 0x40 0x48 0x50 0x58 0x60

Quiz 6: What should be printed?

QUIZ: What should `print(0x21)` show?

(A) (0, (1, (2, false)))
(B) (3, (4, (5, false)))
(C) (0, (1, (2, (3, (4, (5, false))))))
(D) (3, (4, (5, (0, (1, (2, false))))))
(E) (2, (1, (0, (3, (4, (5, false))))))

2	0x51	1	0x01	0	0x11	5	false	4	0x31	3	0x41
---	------	---	------	---	------	---	-------	---	------	---	------

0x00 0x08 0x10 0x18 0x20 0x28 0x30 0x38 0x40 0x48 0x50 0x58 0x60

Quiz 7: Your turn!

What is something you found confusing in today’s lecture (or earlier)?

NAME

CSE 231 SPRING 2026

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